

# Computer Networks: TCP vs UDP

## Transport Layer Overview

The transport layer is responsible for delivering data between applications running on different hosts across a network. The two most widely used transport layer protocols on the internet are the Transmission Control Protocol (TCP) and the User Datagram Protocol (UDP). Both sit on top of the Internet Protocol (IP), but they offer very different guarantees to the applications that use them.

## TCP: Reliable, Connection-Oriented Delivery

TCP establishes a connection between sender and receiver before any data is transmitted, using a process known as the three-way handshake. TCP guarantees that data arrives reliably, in order, and without duplication. It achieves this using sequence numbers, acknowledgments, and retransmission of lost segments.

TCP also implements flow control, to prevent a fast sender from overwhelming a slow receiver, and congestion control, to prevent the sender from overwhelming the network itself. These mechanisms make TCP well suited for applications where correctness matters more than speed, such as file transfer, email, and web browsing.

## UDP: Fast, Connectionless Delivery

UDP is a much simpler protocol. It does not establish a connection, does not guarantee delivery, does not guarantee ordering, and does not perform congestion control. A UDP sender simply sends packets, called datagrams, without waiting for any acknowledgment from the receiver.

Because of this simplicity, UDP has very low overhead and low latency compared to TCP. This makes it a good fit for applications where speed matters more than perfect reliability, such as live video streaming, online gaming, and voice-over-IP calls, where a dropped packet is less harmful than a delayed one.

## Choosing Between TCP and UDP

The choice between TCP and UDP depends entirely on the needs of the application. If an application cannot tolerate missing or out-of-order data, such as a banking transaction or a webpage download, TCP is the correct choice despite its extra overhead. If an application can tolerate some data loss in exchange for lower latency, such as a live video call, UDP is usually preferred, sometimes with a custom reliability layer built on top by the application itself.